Goals for today

- Complete differential kinematics
- Paper review: *Introduction to Inverse Kinematics with Jacobian Transpose, Pseudoinverse and Damped Least Squares methods* [Eva, Antara, & Aswin]*
- Discuss other important uses of the Jacobian (e.g. statics, manipulability, etc.)

*We’ll start inverse kinematics on Monday

General form

For the general case, v_{tip} is a six-dimensional twist $\mathcal{V}$

- $J_i(\theta)$ is the twist $\mathcal{V}$ when $\dot{\theta}_i = 1$ and all other joint velocities $\dot{\theta}$ are zero
- The screw axes of the Jacobian depend on the current joint variables θ (for forward kinematics, the screw axes were always for the case $\theta = 0$)
- The space Jacobian $J_s(\theta)$ satisfies $\mathcal{V}_s = J_s(\theta)\dot{\theta}$ where each column $J_{si}(\theta)$ corresponds to a screw axis expressed in the fixed space frame $\{\mathrm{s}\}$
- The body Jacobian $J_b(\theta)$ satisfies $\mathcal{V}_b = J_b(\theta)\dot{\theta}$ where each column $J_{bi}(\theta)$ corresponds to a screw axis expressed in the body frame $\{\mathrm{b}\}$

Review of twists

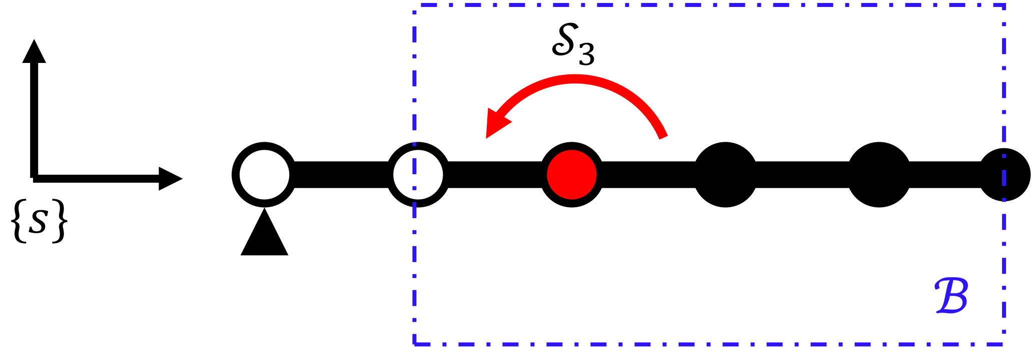
- $\mathcal{V}_s = [\omega_s, v_s]^T$ is the spatial twist, or the spatial velocity, in the space frame
  - ω_s is the angular velocity expressed in $\{\mathrm{s}\}$
  - v_s is the linear velocity of a point at the origin of $\{\mathrm{s}\}$ expressed in $\{\mathrm{s}\}$
- $\mathcal{V}_b = [\omega_b, v_b]^T$ is the body twist, or the spatial velocity, in the body frame
  - ω_b is the angular velocity expressed in $\{\mathrm{b}\}$
  - v_b is the linear velocity of a point at the origin of $\{\mathrm{b}\}$ expressed in $\{\mathrm{b}\}$

Visualizing the space Jacobian (1)

Consider the 5R planar arm to the right. Let $\mathcal{S}_3$ be the screw for joint 3 in the robot's zero configuration.

We know that $\theta_3, \theta_4, \theta_5$ won't affect the spatial twist from $\dot{\theta}_3$ since they won't displace axis 3 in $\{s\}$.

Fix those joint variables at zero, making the robot from joint 2 outward a rigid body $\mathcal{B}$.

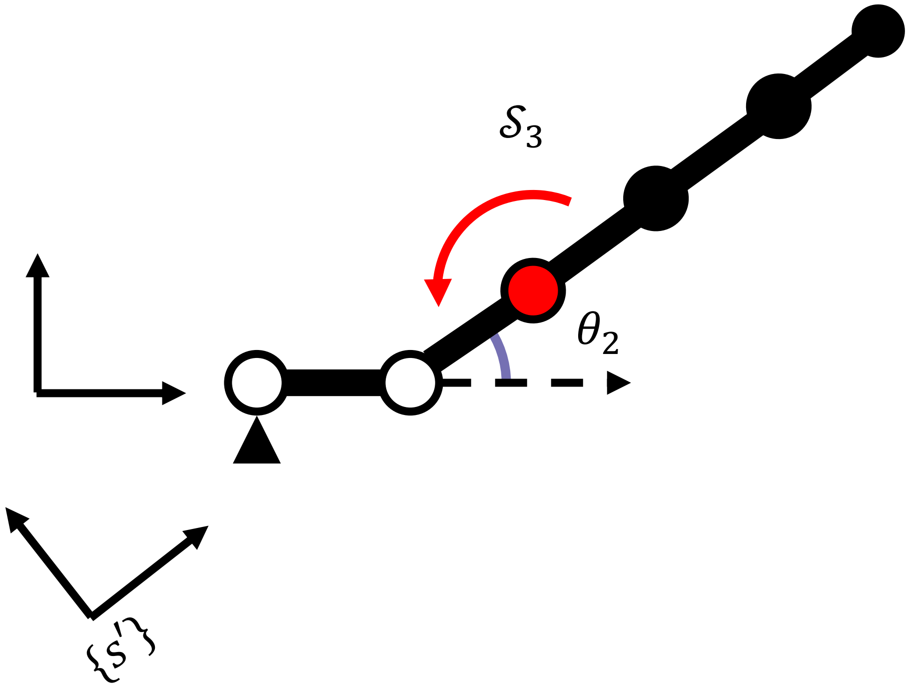


Visualizing the space Jacobian (2)

Now consider when $\theta_1 = 0$ and θ_2 is arbitrary.

Frame $\{s'\}$ is at the same position and orientation relative to $\mathcal{B}$ as frame $\{s\}$ when $\theta_1 = \theta_2 = 0$.

$$T_{ss'} = e^{[\mathcal{S}_2]\theta_2}$$



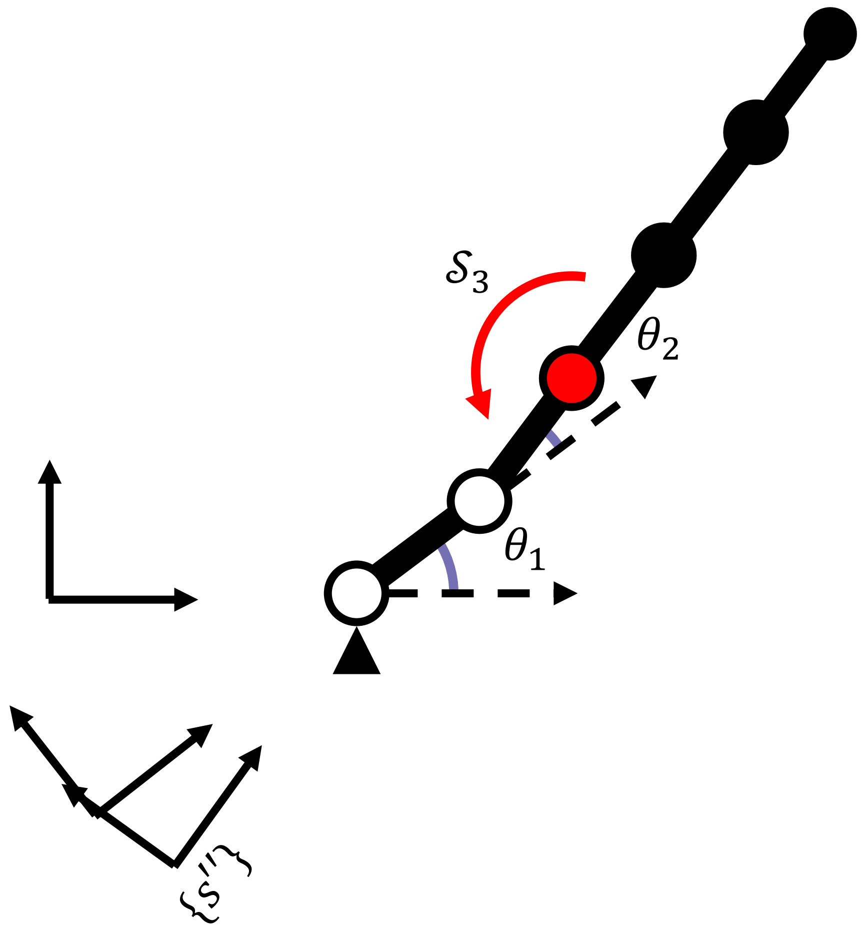
Visualizing the space Jacobian (3)

Now consider when both θ_1 and θ_2 are arbitrary.

Frame $\{s''\}$ is at the same position and orientation relative to $\mathcal{B}$ as frame $\{s\}$ when $\theta_1 = \theta_2 = 0$.

$$T_{ss''} = e^{[\mathcal{S}_1]\theta_1} e^{[\mathcal{S}_2]\theta_2}$$

So, $\mathcal{S}_3$ is the screw relative to $\{s''\}$ for arbitrary joint angles θ_1 and θ_2.



Visualizing the space Jacobian (4)

If we look at the space Jacobian though, the column J_{S3} is the screw relative to $\{\mathrm{s}\}$. How can we relate $\mathcal{S}_3$ to J_{S3}?

We can use adjoint mapping:

$$\left[\mathrm{Ad}_{T_{ss''}}\right] = \left[\mathrm{Ad}_{\mathrm{e}^{[\mathcal{S}_1]\theta_1}\mathrm{e}^{[\mathcal{S}_2]\theta_2}}\right]$$
$$J_{s3} = \left[\mathrm{Ad}_{T_{ss''}}\right]\mathcal{S}_3$$

This way, we can change the representation of $\mathcal{S}_3$ from $\{\mathrm{s}''\}$ to $\{\mathrm{s}\}$.

The space Jacobian (1)

To derive the relationship between an open chain's join velocity vector $\dot{\theta}$ and the end-effector's spatial twist $\mathcal{V}_s$, we first recall a few properties from linear algebra:

i. If $A, B \in \mathbb{R}^{n\times n}$ are both invertible, then $(AB)^{-1} = B^{-1}A^{-1}$

ii. If $A \in \mathbb{R}^{n\times n}$ is constant and $\theta(t)$ is a scalar function of t, then

$$\frac{d\left(\mathrm{e}^{A\theta}\right)}{dt} = A\mathrm{e}^{A\theta}\dot{\theta} = \mathrm{e}^{A\theta}A\dot{\theta}$$

iii. $\left(\mathrm{e}^{A\theta}\right)^{-1} = \mathrm{e}^{-A\theta}$

The space Jacobian (2)

Consider the forward kinematics of an n-link open chain expressed using the PoE formula:

$$T(\theta_1, \dots \theta_n) = \mathrm{e}^{[\mathcal{S}_1]\theta_1}\mathrm{e}^{[\mathcal{S}_2]\theta_2} \dots \mathrm{e}^{[\mathcal{S}_n]\theta_n} M$$

We can relate this to the spatial twist to derive the space Jacobian. Let's start with the spatial twist equation:

$$[\mathcal{V}_s] = \dot{T}T^{-1} = \begin{bmatrix} \dot{R}R^T & \dot{p} - \dot{R}R^T p \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} [\omega_s] & v_s \\ 0 & 0 \end{bmatrix}$$

Using the product rule to find the time derivative of the forward kinematics:

$$\dot{T} = \left(\frac{d}{dt}\mathrm{e}^{[\mathcal{S}_1]\theta_1}\right)\mathrm{e}^{[\mathcal{S}_2]\theta_2} \cdots \mathrm{e}^{[\mathcal{S}_n]\theta_n} M + \mathrm{e}^{[\mathcal{S}_1]\theta_1}\left(\frac{d}{dt}\mathrm{e}^{[\mathcal{S}_2]\theta_2}\right) \cdots \mathrm{e}^{[\mathcal{S}_n]\theta_n} M + \cdots$$

The space Jacobian (3)

$$\dot{T} = \left(\frac{d}{dt}\mathrm{e}^{[\mathcal{S}_1]\theta_1}\right)\mathrm{e}^{[\mathcal{S}_2]\theta_2}\cdots\mathrm{e}^{[\mathcal{S}_n]\theta_n}M + \mathrm{e}^{[\mathcal{S}_1]\theta_1}\left(\frac{d}{dt}\mathrm{e}^{[\mathcal{S}_2]\theta_2}\right)\cdots\mathrm{e}^{[\mathcal{S}_n]\theta_n}M + \cdots$$

Using property (*ii*), we can simplify:

$$\dot{T} = \left([\mathcal{S}_1]\dot{\theta}_1\mathrm{e}^{[\mathcal{S}_1]\theta_1}\right)\mathrm{e}^{[\mathcal{S}_2]\theta_2}\cdots\mathrm{e}^{[\mathcal{S}_n]\theta_n}M + \mathrm{e}^{[\mathcal{S}_1]\theta_1}\left([\mathcal{S}_2]\dot{\theta}_2\mathrm{e}^{[\mathcal{S}_2]\theta_2}\right)\cdots\mathrm{e}^{[\mathcal{S}_n]\theta_n}M + \cdots$$

Using properties (*i*) and (*iii*), we also know:

$$T^{-1} = M^{-1}\mathrm{e}^{-[\mathcal{S}_n]\theta_n}\ldots\mathrm{e}^{-[\mathcal{S}_1]\theta_1}$$

Calculating $\dot{T}T^{-1}$, we obtain:

$$\begin{aligned}&[\mathcal{V}_s] = \dot{T}T^{-1}\\ &= [\mathcal{S}_1]\dot{\theta}_1 + \mathrm{e}^{[\mathcal{S}_1]\theta_1}[\mathcal{S}_2]\mathrm{e}^{-[\mathcal{S}_1]\theta_1}\dot{\theta}_2 + \mathrm{e}^{[\mathcal{S}_1]\theta_1}\mathrm{e}^{[\mathcal{S}_2]\theta_2}[\mathcal{S}_3]\mathrm{e}^{-[\mathcal{S}_2]\theta_2}\mathrm{e}^{-[\mathcal{S}_1]\theta_1}\dot{\theta}_3 + \cdots\end{aligned}$$

The space Jacobian (4)

We can make this more compact by putting it into vector form and using adjoint mapping (*see slides 23-24 from the rigid body motion lecture*):

$$\mathcal{V}_s = \mathcal{S}_1\dot{\theta}_1 + \mathrm{Ad}_{e^{[\mathcal{S}_1]\theta_1}}(\mathcal{S}_2)\dot{\theta}_2 + \mathrm{Ad}_{e^{[\mathcal{S}_1]\theta_1}e^{[\mathcal{S}_2]\theta_2}}(\mathcal{S}_3)\dot{\theta}_3 + \cdots$$

Recall, for any $\mathcal{V} \in \mathbb{R}^6$, the adjoint map associated with T is

$$\mathcal{V}' = [\mathrm{Ad}_T]\mathcal{V} = \mathrm{Ad}_T(\mathcal{V}) \text{ where } [\mathrm{Ad}_T] = \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix} \in \mathbb{R}^{6\times 6}$$

In summary, $\mathcal{V}_s$ is a sum of n spatial twists of the form

$$\mathcal{V}_s = J_{s1}\dot{\theta}_1 + J_{s2}(\theta)\dot{\theta}_2 \ldots + J_{sn}(\theta)\dot{\theta}_n$$

where each $J_{si}(\theta) = \left(\omega_{si}(\theta), v_{si}(\theta)\right)$ depends on the joint values $\theta \in \mathbb{R}^n$ for $i = 2, 3, \ldots, n$.

The space Jacobian (5)

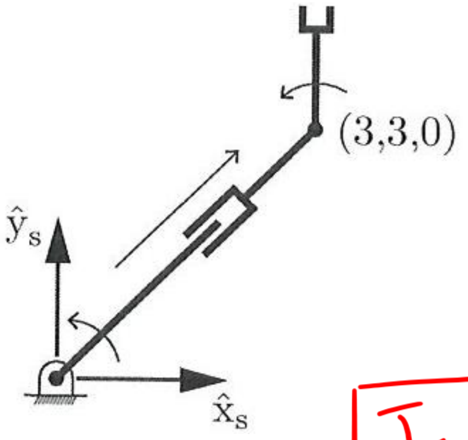
In matrix form, we have:

$$\begin{aligned}\mathcal{V}_s &= [J_{s1}\ \ J_{s2}(\theta) \cdots J_{sn}(\theta)] \begin{bmatrix} \dot{\theta}_1 \\ \vdots \\ \dot{\theta}_n \end{bmatrix} \\ &= J_s(\theta)\dot{\theta}\end{aligned}$$

$J_s(\theta)$ is the **space Jacobian**, or Jacobian in fixed frame coordinates.

The i^{th} column $J_{si}(\theta)$ of $J_s(\theta)$ is the screw axis describing joint axis i, expressed in fixed frame coordinates, as a function of joint variables $\theta_1, \theta_2, \ldots, \theta_{i-1}$.

Example



A planar RPR robot, shown below, has a frame $\{s\}$ at joint 1. The robot moves in the plane $z_s = 0$. Joint 3 is at the point $(3, 3, 0)$ in $\{s\}$. The direction of positive motion at each joint is shown. The end-effector twist satisfies $\mathcal{V}_s = J_s(\theta)\dot{\theta}$. $J_s(\theta)$ is a 6 x 3 matrix, since the twist is six-dimensional and there are only three joints.

Write the 6 x 3 space Jacobian $J_s(\theta)$ at the configuration shown.

$\boxed{J_{s_1}}$

$\omega_{s_1} = [0 \quad 0 \quad 1]^T$

$v_{s_1} = [0 \quad 0 \quad 0]^T$

$\boxed{J_{s_2}}$

$\omega_{s_2} = [0 \quad 0 \quad 0]^T$

$v_{s_2} \Rightarrow$

v_{s_2}

$45°$

$v_{s_2} = [\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0]^T$

$\boxed{J_{s_3}}$

$\omega_{s_3} = [0 \quad 0 \quad 1]^T$

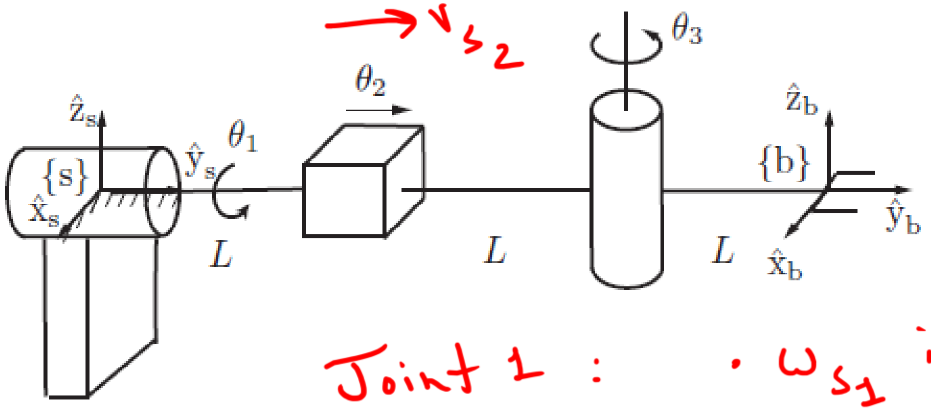
$q_3 = [3 \quad 3 \quad 0]^T$

$$v_{s_3} = -\omega_{s_3} \times q_3 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & -1 \\ 3 & 3 & 0 \end{vmatrix} = \hat{i}(3) - \hat{j}(3) + \hat{k}(0)$$

$$J_s = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1/\sqrt{2} & 3 \\ 0 & 1/\sqrt{2} & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

at the configuration shown

Example



The RPR robot is shown in its zero position. Find the space Jacobian $J_s(\theta)$ for arbitrary configurations $\theta \in \mathbb{R}^3$.

Joint 1 :

- ω_{s_1} is constant in the $\hat{y}_s$ direction

$\omega_{s_1} = [0 \;\; 1 \;\; 0]^T$

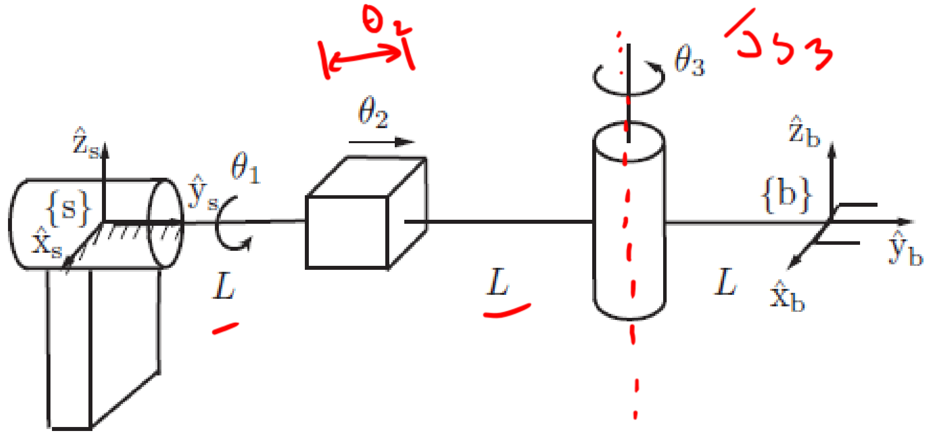
- choose q_1 as the origin $(0,0,0)$

$v_{s_1} = [0 \;\; 0 \;\; 0]^T$

Joint 2 : A prismatic joint

$\omega_{s_2} = [0 \;\; 0 \;\; 0]^T$

$$v_{s_2} = \mathrm{Rot}(\hat{y}_s, \theta_1)\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} c\theta_1 & 0 & s\theta_1 \\ 0 & 1 & 0 \\ -s\theta_1 & 0 & c\theta_1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$



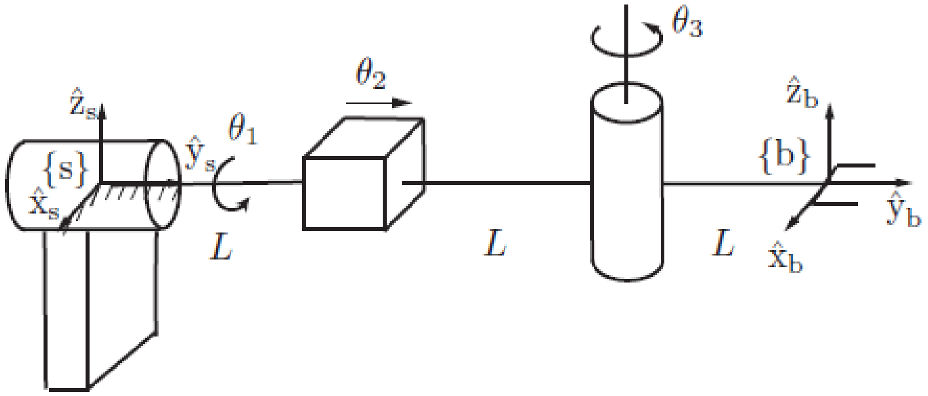
Joint 3 :

$$\omega_{s_3} = \begin{bmatrix} c\theta_1 & 0 & s\theta_1 \\ 0 & 1 & 0 \\ -s\theta_1 & 0 & c\theta_1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} s\theta_1 \\ 0 \\ c\theta_1 \end{bmatrix}$$

$$v_{s_3} = -\omega_{s_3} \times q_3$$

$$q_3 = \begin{pmatrix} 0 & (2L+\theta_2) & 0 \end{pmatrix}$$

$$= \begin{bmatrix} 0 & c\theta_1 & 0 \\ -c\theta_1 & 0 & s\theta_1 \\ 0 & -s\theta_1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 2L+\theta_2 \\ 0 \end{bmatrix} = \begin{bmatrix} c\theta_1(2L+\theta_2) \\ 0 \\ -s\theta_1(2L+\theta_2) \end{bmatrix}^T$$



$$J_s(\theta) = \begin{bmatrix} 0 & 0 & \sin\theta_1 \\ 1 & 0 & 0 \\ 0 & 0 & \cos\theta_1 \\ 0 & 0 & \cos\theta_1(2L+\theta_2) \\ 0 & 1 & 0 \\ 0 & 0 & -\sin\theta_1(2L+\theta_2) \end{bmatrix}$$

$$J_s(\theta) = \begin{bmatrix} 0 & 0 & s\theta_1 \\ 1 & 0 & 0 \\ 0 & 0 & c\theta_1 \\ 0 & 0 & c\theta_1(2L+\theta_2) \\ 0 & 1 & 0 \\ 0 & 0 & -s\theta_1(2L+\theta_2) \end{bmatrix}$$

The adjoint matrices are implicit in our calculations of the screw axes of the displaced joint axes

Using the adjoint representation, which is more efficient for general computation purposes (set L = 1 and theta = [pi/2, 2, pi/2]^T)

```
S_home = [0 0 0; 1 0 0; 0 0 1; 0 0 2; 0 1 0; 0 0 0];
theta = [pi/2 2 pi/2];
sJacob(S_home,theta)

ans =

         0         0    1.0000
    1.0000         0         0
         0         0    0.0000
         0         0    0.0000
         0    1.0000         0
         0         0   -4.0000
```

$$J_s(\theta) = \begin{bmatrix} 0 & 0 & \sin\theta_1 \\ 1 & 0 & 0 \\ 0 & 0 & \cos\theta_1 \\ 0 & 0 & \cos\theta_1(2L+\theta_2) \\ 0 & 1 & 0 \\ 0 & 0 & -\sin\theta_1(2L+\theta_2) \end{bmatrix}$$

Which matches what we found directly

$\sin(\pi/2) = 1$ ✓
0 ✓
$\cos(\pi/2) = 0$ ✓
0 ✓
0 ✓
-4 ✓

```
function Js = sJacob(S,theta)
% Calculates the space Jacobian
% Inputs: S - 6 x n matrix of screw axes in the zero configuration
%         theta - vector of n joint coordinates
% Output: Js - 6 x n space Jacobian matrix

c = length(S(1,:)); % # of columns in S
Js(:,1) = S(:,1);   % first column of Js is just S1
for i = 2:c
    A = eye(6);      % initialize a 6 x 6 identity matrix
    for j = 1:i-1
        B = MatExp(S(:,j),theta(j)); % find SE(3)
        B = adjointM(B);             % adjoint representation of SE(3)
        C = A*B;
        A = C;      % iterate until joint i-1
    end
    Js(:,i) = C*S(:,i);
end
```

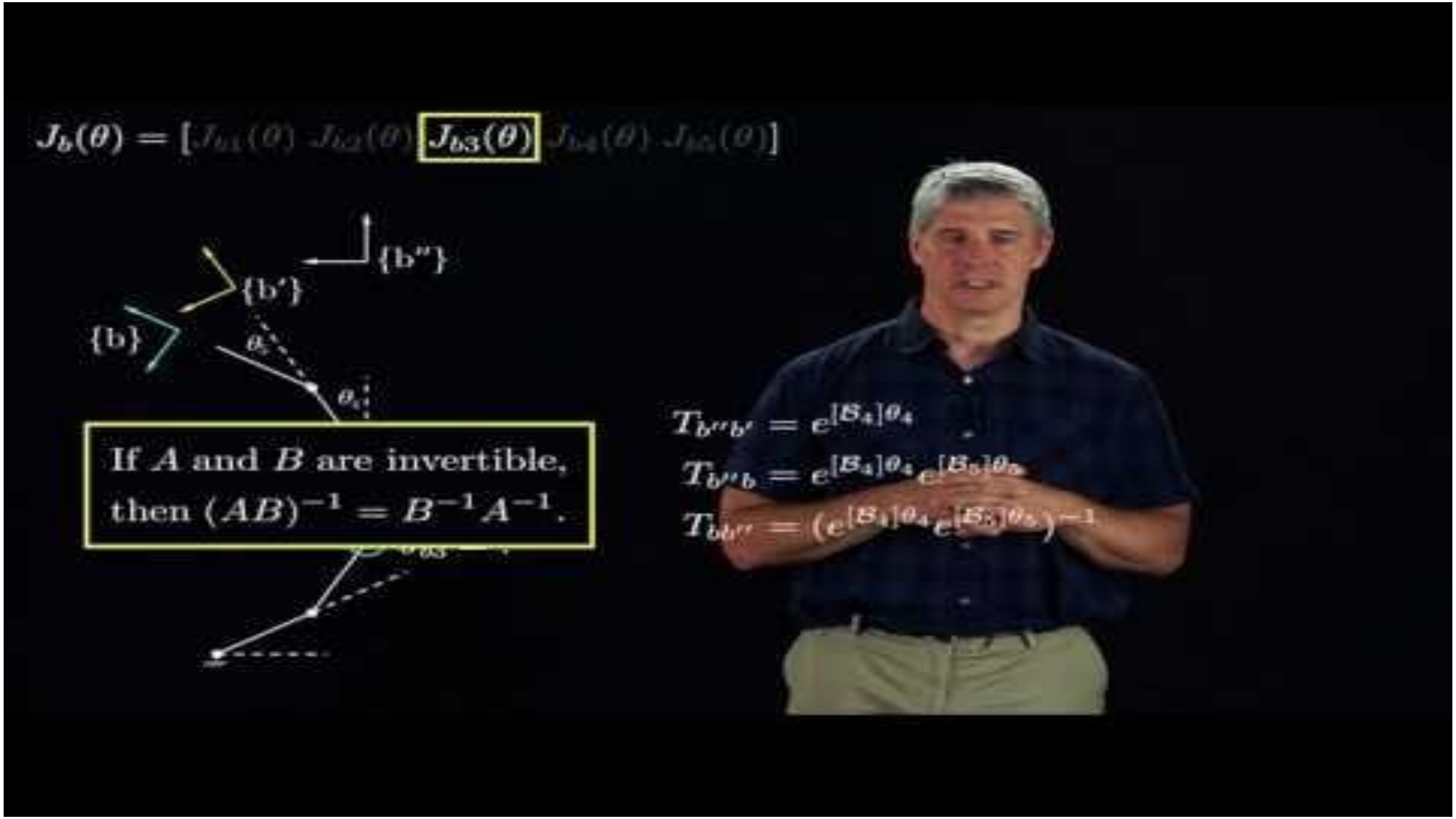
```
function A = adjointM(T)
% Returns the adjoint map associated with an SE(3)
% Inputs: T - homogeneous transformation matrix SE(3)
% Output: A - 6 x 6 adjoint representation

% See equation 3.76
A(1:3,1:3) = T(1:3,1:3);
A(1:3,4:6) = zeros;
A(4:6,1:3) = skew(T(1:3,4))*T(1:3,1:3);
A(4:6,4:6) = T(1:3,1:3);
```

```
function T = MatExp(S,theta)
% Calculates the homogeneous transformation for a set of exponential
% coordinates
% Inputs: S - screw axis; theta - distance along the screw axis
% Output: T - homogeneous transformation matrix SE(3)

if sum(S(1:3)) == 0 % check if angular velocity is zero
    type = 'P';      % prismatic joint
else
    type = 'R';      % revolute joint
end

wSkew = skew(S(1:3));    % skew-symmetric representation
v = [S(4), S(5), S(6)]';

switch type
    case 'R' % Revolute joint (See eqn. 3.88)
        R = eye(3)+sin(theta)*wSkew+(1-cos(theta))*wSkew^2;
        p = (eye(3)*theta+(1-cos(theta))*wSkew+(theta-sin(theta))*wSkew^2)*v;
    case 'P' % Prismatic joint (See eqn. 3.89)
        R = eye(3);
        p = v*theta;
end
T(1:3,1:3) = R;
T(:,4) = p;
T(4,:) = [0 0 0 1];
```

The i[th] column of $J_s(\theta) \in \mathbb{R}^{6\times n}$ is given by

$$J_{si}(\theta) = Ad_{e^{[S_1]\theta_1}\cdots e^{[S_{i-1}]\theta_{i-1}}}(S_i) \text{ for } i = 2, \dots, n \text{ with the first column } J_{s1} = S_1$$

The body Jacobian



The body Jacobian

The body Jacobian $J_b(\theta) \in \mathbb{R}^{6\times n}$ relates the joint velocities $\dot{\theta} \in \mathbb{R}^n$ to the end effector twist $\mathcal{V}_b = [\omega_b \quad v_b]$ via

$$\mathcal{V}_b = J_b(\theta)\dot{\theta}$$

The i^{th} column of $J_b(\theta)$ is $J_{bi}(\theta) = \mathrm{Ad}_{\mathrm{e}^{-[B_n]\theta_n}\dots\mathrm{e}^{-[B_{i+1}]\theta_{i+1}}}(B_i)$ for $i = n-1, \dots, 1$ with $J_{bn} = B_n$.

The i^{th} column $J_{bi}(\theta)$ of $J_b(\theta)$ is the screw axis describing joint axis i, expressed in the coordinates of the end-effector frame.